Learning When and Where to Handover: A Hierarchical Reinforcement Learning Framework for Dense LEO Satellite Constellations

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***Abstract*—The rapid development of low Earth orbit (LEO) satellite constellations has enabled global broadband coverage and low-latency communication, but also brings new challenges to connection continuity and network stability. Due to the ultra-dense deployment and high orbital velocity of LEO satellites, users frequently experience mandatory handovers and are prone to triggering "ping-pong" effects in overlapping coverage areas. To address these issues, we propose a hierarchical reinforcement learning (HRL)-based inter-satellite handover framework that decouples decision-making into two subproblems: when to initiate a handover at the temporal level and which satellite to handover to at the spatial level. To achieve long-term planning, the temporal agent utilizes the proximal policy optimization (PPO) algorithm to determine the handover timing by predicting future satellite load and interference levels. For short-term adaptability, the spatial agent employs the deep Q-network (DQN) to select the target satellite based on a utility function incorporating satellite load, signal quality, and service duration. We implement the proposed method in a Starlink constellation environment. Simulation results show that the proposed HRL-based scheme outperforms traditional and non-hierarchical RL methods in terms of user handover frequency and handover success rate. Furthermore, we validate that the proposed approach significantly improves transmission rate and load balancing, demonstrating its effectiveness in highly dynamic LEO satellite environments.**

***Index Terms*—Dense LEO satellite networks, inter-satellite handover, inter-satellite interference, hierarchical reinforcement learning, satellite load balance.**

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I. Introduction

THE development of 6G technology is driving a growing demand for global seamless network coverage and ultra-low latency communication. The traditional terrestrial communication networks are constrained by geographical limitations, leaving coverage gaps in oceans, deserts, and remote areas [1], [2], [3]. The geostationary Earth orbit (GEO) satellites can achieve wide-area coverage, but their high orbit causes significant latency, making them unsuitable for real-time interactive services. In contrast, low Earth orbit (LEO) satellites, due to their low latency, extensive coverage, and flexible deployment capabilities, have emerged as a pivotal technology for achieving global seamless communication [4]. Commercial aerospace companies like SpaceX's Starlink [5] and OneWeb [6] are rapidly expanding their LEO constellations, pushing LEO satellite communication systems toward ultra-large-scale and ultra-high-density developments.

However, the ultra-dense deployment and high-speed movement of LEO satellites also pose significant challenges to the continuity and stability of user connections. Specifically, there are two major technical challenges.

- Due to the high orbital velocity of LEO satellites, the visibility period for ground users is relatively short. To ensure continuous services, users must undergo mandatory handovers when exiting the coverage area.
- Due to the high-speed movement of LEO satellites, the signal strength fluctuates rapidly and significantly. If a user terminal is located in the overlapping coverage area of two or more satellites, it is highly prone to triggering "ping-pong" handovers [7].

Given the frequent satellite movements, short visibility durations, and overlapping coverage areas, an adaptive and robust inter-satellite handover mechanism is crucial to ensure stable and high-quality communication.

A. Related Work

In recent years, a lot of research has been carried out for handover strategies in LEO satellite constellations. The traditional methods are mainly based on graph theoretic modeling [8], geometric prediction [9], and quality of service (QoS) threshold [10] triggering mechanisms. Specifically, a weighted

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graph-based handover decision framework was developed by modeling handover operations as edges and satellite coverage areas as nodes [11]. An orbital-geometry-based pre-handover algorithm was proposed in [12], which utilizes satellite ephemeris data to determine the optimal handover timing in advance. The authors in [13] proposed a dynamic handover strategy based on multiple threshold triggers, achieving rapid and accurate handover by jointly evaluating signal strength, satellite load, and other parameters. Similarly, a cross-layer handover strategy that integrates information from both the physical and network layers has been investigated in [14]. The strategy adaptively adjusted the handover trigger conditions by combining real-time channel state information (CSI), routing tables, and service priorities. In [15], a cost-optimized handover mechanism was presented for delay-sensitive internet of things (IoT) services, where the target satellite is dynamically selected by jointly assessing communication latency, computational overhead, and interruption loss. However, as the number of satellites increases, the performance of the above handover schemes gradually degrades. The main reasons are as follows. I)***The coupling of multiple constraints:*** Traditional handover strategies often involve multiple heterogeneous constraints, such as signal strength, satellite load, and coverage area, which are highly coupled. Due to the high-speed movement of satellites and the frequent and unpredictable changes in environment parameters, handover strategies based on static thresholds or linear weighting mechanisms struggle to provide real-time responses in the highly dynamic LEO networks. II)***The increased computational complexity:*** In large-scale LEO satellite networks, the number of satellites visible to each user increases sharply, significantly expanding the set of potential handover targets. Moreover, due to orbital dynamics, these candidate combinations change continuously over time. Traditional methods rely on global handover path searches [16] or enumerating a vast number of possible handover paths, resulting in exponential computational overhead.

In the face of the complex and dynamic environment of dense LEO satellite networks, artificial intelligence (AI) technologies are emerging as a promising solution. In particular, deep reinforcement learning (DRL), with its capability for continuous interaction and autonomous learning, has demonstrated significant potential in areas such as beam-hopping pattern design [17], multi-resource allocation optimization [18], [19], and user handover decision-making [20]. The authors in [21] utilized DRL to design a multi-objective handover strategy, optimizing system throughput, handover frequency, and load balancing for intra- and inter-satellite scenarios. A multi-agent strategy for user handover was introduced in large-scale satellite constellations [22], leveraging Markov and Loo models to capture channel dynamics and balance network demand and load. In [23], a distributed handover mechanism was proposed based on the multi-agent fingerprint-enhanced double deep Q-network (MAF-DDQN-DIH) algorithm, with the aim of minimizing computational burden. To address the issues of limited storage efficiency, [24] presented a DQN-based handover framework incorporating an adaptive learning rate and a momentum mechanism. A distributed handover framework based on federated learning was developed in [25], where each satellite locally trains a lightweight model and collaboratively aggregates model parameters across the network to enable adaptive and fault-tolerant handover decisions. Although these AI-based schemes have improved satellite handover strategies, they still exhibit some shortcomings. I) ***Homogeneous feature consideration:*** Most current approaches model the inter-satellite handover as a unified process, failing to exploit the distinct temporal (e.g., signal degradation trends) and spatial (e.g., satellite coverage overlaps) characteristics that inherently influence the decision-making process. This will lead to slow convergence and poor adaptability in rapidly evolving LEO satellite networks. II) ***Shortsighted decision-making:*** Existing algorithms typically rely on the current network snapshot to make decisions, overlooking the long-term evolution of system states driven by satellite mobility and user dynamics. This will lead to ineffective handovers and unbalanced loads, ultimately degrading the long-term performance of the LEO networks.

In addition to the above studies, mobility management in non-terrestrial networks (NTNs) has also been extensively investigated in 3GPP standardization activities. Existing NTN specifications mainly focus on protocol and signaling support for mobility [42], including beam management, measurement configuration and reporting, as well as standardized procedures for intra-satellite beam handover and inter-satellite handover, to guarantee access continuity and interoperability under satellite mobility. These works primarily adopt event-based or threshold-based triggering mechanisms and predefined signaling workflows, while the optimization of handover timing and target selection under dense multi-satellite coverage is largely left open for implementation.

B. Motivations and Contributions

To address the challenges faced by the aforementioned solution in ultra-dense LEO satellite networks, this paper proposes an inter-satellite handover scheme based on hierarchical reinforcement learning (HRL). Here, the inter-satellite handover problem is decoupled into two subproblems: **"when to handover"** and **"which to handover"**. On the one hand, the determination of handover timing is a time-sensitive decision, which relies on the prediction of variations in signal quality, interference levels, and satellite load. On the other hand, the optimization of the handover target is a spatial-selection decision, focusing on the joint evaluation of multiple factors such as channel quality, service duration, and resource occupancy of candidate satellites. Therefore, there are fundamental differences between the two in terms of decision logic and information acquisition. If combined into a single policy model, the learning complexity would increase significantly, resulting in slow convergence and poor stability. By decomposing the task, each sub-policy can concentrate on optimizing within its specific decision domain, effectively reducing the dimensionality of the decision space and simplifying the training process. At the same time, the information exchange mechanism between different decision levels enhances coordination among sub-policies, thereby improving the efficiency and adaptability of the overall handover strategy. The main contributions of this paper are summarized as follows.

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1) We develop a HRL-based approach that explicitly decouples the inter-satellite handover problem into two subproblems, i.e., when to initiate a handover and which satellite to handover to. Unlike existing DRL-based handover schemes, the proposed framework separates long-term handover trigger from short-term target selection, significantly reducing decision complexity and improving learning stability in dense LEO satellite networks.
2) To provide long-term planning, the temporal agent is responsible for determining the handover timing based on the proximal policy optimization (PPO) algorithm. We design a gated recurrent unit (GRU)-based actor network to predict future satellite load and interference levels, while employing a critic network to evaluate whether to initiate a handover.
3) To enable short-term responsiveness, the spatial agent is accountable for selecting the handover target based on the deep Q-network (DQN) algorithm. We construct a utility function that jointly considers signal quality, service duration, and satellite load, and select the optimal target by comparing the Q-values of candidate satellites under diverse coverage conditions.
4) We validated the convergence performance of the HRL-based approach within the Starlink constellation environment. Experimental results demonstrate that compared to traditional methods and non-hierarchical DRL models, the proposed inter-satellite handover scheme achieves superior performance in terms of handover frequency, transmission rate and satellite load balancing.

The rest of the paper is organized as follows. Section II introduces the system model. Section III elaborates on the proposed inter-satellite handover method based on HRL. The simulation results are compared and analyzed in Section IV. Section V concludes the full paper.

II. System Model

In this section, we begin with an overview of the dense LEO satellite network, followed by a detailed description of the channel model and inter-satellite interference. We then define satellite load and develop a prediction model to capture its spatio-temporal dynamics. Finally, we formulate the optimization problem concerning user handover.

A. System Model

As illustrated in Fig. 1, we consider a dense LEO satellite network consisting of M LEO satellites and N ground users. The set of LEO satellites is denoted as $\mathbb{S} = \{s_1, s_2, \ldots, s_M\}$, where each satellite moves along a predetermined orbit at a constant velocity, with an orbital period of T. Moreover, the set of ground users is represented as $\mathbb{U} = \{u_1, u_2, \ldots, u_N\}$, and they are non-uniformly distributed within the coverage region of the satellite network. To characterize the stochastic and bursty nature of traffic demands, service arrivals are modeled as a Poisson process with an arrival rate of λ. We define a satellite $s_i \in \mathbb{S}$ to be visible to a user $u_j \in \mathbb{U}$ if the elevation angle $\theta_{i,j}$ between them exceeds a predefined threshold θ_{min}.

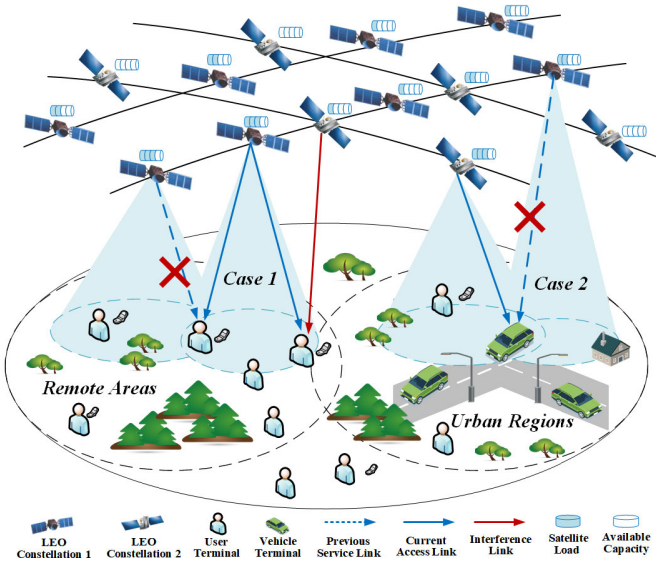


Fig. 1. The illustration of user handover in dense LEO satellite networks. (In Case 1, the handover is passively triggered as the user exits the satellite coverage area. In Case 2, the handover is actively initiated by the user in response to degraded service quality.)

In dense LEO constellation scenarios, the coverage areas of multiple satellites may overlap, allowing a single user to be simultaneously visible to multiple satellites. Consequently, the set of visible satellites for user u at time t is denoted by $\mathbf{V}_u(t) \subseteq \mathbb{S}$.[1] At any time, each user is allowed to establish a communication link with only one LEO satellite [26], i.e., $\sum_{s\in\mathbb{S}} x_{u,s}(t) = 1$, where $x_{u,s}(t) \in \{0, 1\}$ denotes whether user u is associated with satellite s at time t. This multi-satellite visibility introduces flexibility for access optimization, but also leads to frequent inter-satellite handovers.

Due to the highly dynamic network topology caused by satellite motion, the visibility set $\mathbf{V}_u(t)$ changes rapidly over time, forcing users to frequently reselect the serving satellites. Therefore, it is essential to design an efficient inter-satellite handover decision mechanism that enables each user to select the optimal satellite from its visible set $\mathbf{V}_u(t)$ at each time instant. This decision-making process should jointly consider time-varying service demands, channel conditions, satellite load, and handover cost, aiming to ensure seamless connectivity while achieving efficient data transmission.

B. Channel Model and Interference Analysis

In this paper, the channel between a ground user $u \in \mathbb{U}$ and a LEO satellite $s \in \mathbb{S}$ at time t is affected by several propagation phenomena, including large-scale path loss, shadowing, and small-scale fading.

The free space path loss can be expressed as

$$H_{u,s}(t) = \left(\frac{4\pi d_{u,s}(t) f_c}{c}\right)^2, \tag{1}$$

[1]The visible satellite set $\mathbf{V}_u(t)$ is updated at every time slot based on the instantaneous satellite–user geometry. This update is synchronized with the base simulation time step and is independent of the decision frequencies of the temporal and spatial agents.

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where $d_{u,s}$ denotes the distance between the user u and the satellite s at time t, and f_c is the carrier frequency. Moreover, shadow fading is a slowly varying random attenuation caused by large obstacles. This can be modeled as a log-normal random variable $F_{u,s}(t)$ with zero mean and a certain variance, represented by $10\log_{10} F_{u,s}(t) \sim \mathcal{N}\left(0, \sigma^2\right)$.

Given the relatively high altitude of LEO satellites, the line-of-sight (LOS) component typically dominates the channel characteristics. Therefore, it is modeled as a Rician random variable $M_{u,s}(t)$, which follows a Rician distribution with factor K [27], denoted by $M_{u,s}(t) \sim \text{Rician}(K)$.

Considering the above factors, the channel gain $h_{u,s}(t)$ between user u and satellite s can be modeled as

$$h_{u,s}(t) = \frac{\sqrt{G_t G_r}c}{4\pi d_{u,s}(t) f_c}\sqrt{F_{u,s}(t)}M_{u,s}(t), \tag{2}$$

where G_t and G_r are the transmitter and receiver antenna gains, respectively. In existing studies, the intra-satellite interference between adjacent beams has been well studied and managed using techniques like spectrum coordination [28], [29] and power control [30], [31]. However, in dense LEO constellations, overlapping coverage from multiple satellites will cause significant inter-satellite interference. Suppose user u is connected to satellite s at time t. Thus, the inter-satellite interference experienced by user u can be expressed as

$$I_{u,s}(t) = \sum_{v \in V_u(t)/s} P_v(t)|h_{u',v}(t)|^2, \tag{3}$$

where $P_v(t)$ denotes the transmit power of the interfering satellite v at time t, and $h_{u',v}(t)$ represents the channel gain between the interfering satellite v and other covered users.

According to the Shannon formula, the transmission rate of user u is calculated based on the signal to interference plus noise ratio (SINR), which is expressed by

$$R_{u,s}(t) = B_{u,s}(t)\log_2\left(1 + \frac{P_s(t)|h_{u,s}(t)|^2}{I_{u,s}(t) + N_0}\right). \tag{4}$$

Here, $P_s(t)$ represents the transmit power, $B_{u,s}(t)$ denotes the bandwidth allocated to user u at time t, and N_0 is the noise power. It is worth noting that, in high-mobility satellite communication scenarios, advanced waveform techniques such as orthogonal time frequency space (OTFS) modulation have been widely recognized as effective solutions for mitigating severe Doppler effects and enhancing channel robustness [41]. In this work, we assume that such Doppler-resilient waveforms are adopted at the physical layer, so that reliable channel quality indicators can be observed and utilized for handover decision-making at the network level.

C. Factor Modeling and Load Prediction

In LEO satellite networks, the satellite load reflects the utilization of satellite communication resources. Considering the total traffic demand and the satellite capacity, the normalized load of satellite s at time t is defined as

$$L_s(t) = \frac{D_s(t)}{C_s}. \tag{5}$$

Here, $D_s(t)$ indicates the amount of service demand carried by satellite s at time t, and C_s is the total service capacity.

In addition, the satellite load exhibits strong temporal and spatial variation, which is influenced by: I) the number and distribution of users in its footprint, II) the volume and QoS requirements of ongoing user sessions, and III) the time-varying handover inflow/outflow due to orbital motion. Here, we aim to model and predict the future load of each satellite by jointly considering traffic dynamics, satellite mobility, and historical behavior of satellite loads. Specifically, the modeling involves the following components.

- **User Traffic Modeling:** The arrival of service requests is modeled as a Poisson process with the arrival rate λ_u. Moreover, the duration of each service session is assumed to follow an exponential distribution [32], represented by a random variable τ_u. Combining the instantaneous data rate $R_{u,s}(t)$, the expected traffic contribution of user u to LEO satellite s is represented as

$$\bar{D}_{u,s}(t) = \lambda_u \mathbb{E}\left[\tau_u\right] R_{u,s}(t). \tag{6}$$

  Furthermore, the traffic demand within the coverage area of satellite s can be modeled as

$$D_s(t) = \iint_{(x,y)\in O_s(t)} \bar{D}_{u,s}(x,y,t)\rho_u(x,y,t)\mathrm{d}x\mathrm{d}y, \tag{7}$$

  where $O_s(t)$ is the footprint area of satellite s at time t, and $\rho_u(x,y,t)$ is the non-station and time-varying spatial user density function.
- **Satellite Mobility Modeling:** First, the future trajectory of satellite s can be computed based on orbital mechanics using methods such as Two-Line Element (TLE) parameters [33]. Then, the visibility period between satellite s and user u is predicted based on their relative positions. The visibility condition is defined as

$$\mathbb{I}_{u,s}(t) = \begin{cases} 1, & \text{if } \theta_{u,s}(t) \geq \theta_{min}, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

  Furthermore, this visibility window is critical for determining when a user enters or exits the LEO satellite's coverage area. Accordingly, the load variation of satellite s in interval $[t, t+\Delta t]$ can be represented as

$$\Delta D_s^{\text{in}}(t) = \sum_{\Phi_{in}} \mathbb{I}_{u,s}(t+\Delta t)\bar{D}_{u,s}(t), \tag{9a}$$

$$\Delta D_s^{\text{out}}(t) = \sum_{\Phi_{out}} \left(1 - \mathbb{I}_{u,s}(t+\Delta t)\right)\bar{D}_{u,s}(t). \tag{9b}$$

  Here, $\Phi_{in} = \{u | u \notin \mathcal{U}_s(t), u \in \mathcal{U}_s(t+\Delta t)\}$ and $\Phi_{out} = \{u | u \in \mathcal{U}_s(t), u \notin \mathcal{U}_s(t+\Delta t)\}$ jointly characterize the time-varying user association dynamics induced by satellite motion and user mobility.

In this work, we model the satellite load by formulating it as a function of historical load states and user inflow/outflow traffic. Given that satellite load evolves over time, the load state of satellite s can be modeled as a time series

$$\mathcal{L}_s = \{L_s(t_0), L_s(t_1), \ldots, L_s(T)\}. \tag{10}$$

If the satellite load is influenced by additional factors [34], we construct a multi-variate time series model, formulated by

$$L_s(t+\Delta t) = \mathcal{F}\left(L_s(t-k\Delta t : t), \mathbf{z}_s(t-k\Delta t : t)\right). \tag{11}$$

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Here, $\mathcal{F}(\cdot)$ is the prediction function, and $\mathbf{z}_s$ represents external features such as user density, traffic rate, or number of visible satellites. The parameter k controls the size of the time window. Therefore, users should consider both current and future load conditions to avoid connecting to potentially overloaded satellites, while load balancing must also be achieved to improve overall network stability.

D. Problem Formulation

In this paper, our goal is to maximize the system transmission rate, achieve satellite load balancing, and minimize unnecessary handovers as much as possible to ensure service continuity and network stability. Specifically, the system throughput at time t is expressed as

$$R(t) = \sum_{s\in\mathbb{S}} \sum_{u\in\mathbb{U}} x_{u,s}(t) R_{u,s}(t). \tag{12}$$

Moreover, the average variance of satellite loads at time t is defined as the load balancing metric [35], represented by

$$E(t) = \frac{1}{M} \sum_{s\in\mathbb{S}} \left(L_s(t) - \bar{L}(t)\right)^2, \tag{13}$$

where $\bar{L}(t)$ is the average satellite load at time t. To determine whether user u performs a handover at time t, we compare the current and previous association states. Then, the number of handovers in the system at time t can be calculated as

$$Q(t) = \frac{1}{2} \sum_{u\in\mathbb{U}} \sum_{s\in\mathbb{S}} \left| x_{u,s}(t) - x_{u,s}(t-1) \right|. \tag{14}$$

Therefore, the optimization problem of inter-satellite handover can be formulated as

$$\max_{x_{u,s}(t)} \quad \sum_{t=1}^{T} \left(\alpha R(t) - \beta E(t) - \eta Q(t)\right), \tag{15a}$$

$$s.t.\ C1: x_{u,s}(t) \le \mathbb{I}_{u,s}(t),\ \forall u \in \mathbb{U},\ \forall s \in \mathbb{S},\ \forall t, \tag{15b}$$

$$C2: \sum_{s\in\mathbb{S}} x_{u,s}(t) \le 1,\ \forall s \in \mathbb{S},\ \forall t, \tag{15c}$$

$$C3: \sum_{u\in\mathbb{U}} x_{u,s}(t) R_{u,s}(t) \le C_s,\ \forall s \in \mathbb{S},\ \forall t, \tag{15d}$$

$$C4: \sum_{s\in\mathbb{S}} x_{u,s}(t) R_{u,s}(t) \ge R_u^{min},\ \forall u \in \mathbb{U},\ \forall t, \tag{15e}$$

$$C5: \sum_{t\in\mathbb{T}} Q_u(t) \le \Gamma,\ \forall u \in \mathbb{U}, \tag{15f}$$

$$C6: \sum_{u\in\mathbb{U}} x_{u,s}(t) B_{u,s}(t) \le B_s^{total},\ \forall s \in \mathbb{S},\ \forall t, \tag{15g}$$

$$C7: P_s(t) \le P_s^{max},\ \forall s \in \mathbb{S},\ \forall t, \tag{15h}$$

In Eq. (15a), α, β and η are the weighting coefficient. The optimization is subject to the following constraints. Constraint $C1$ is the visibility constraint, indicating that a user can only perform a handover when the LEO satellite is visible. Constraint $C2$ ensures the exclusivity of user-satellite connections. Constraint $C3$ enforces the satellite capacity limitation, while Constraint $C4$ ensures the minimum data rate requirement to maintain fundamental QoS. To mitigate the "ping-pong" effect and signaling overhead, Constraint $C5$ imposes an upper bound on handover frequency during time window T. Furthermore, Constraint $C6$ indicates the bandwidth allocation boundary per satellite, and Constraint $C7$ strictly maintains the maximum transmission power constraint for each satellite.

For the aforementioned complex problem, traditional optimization methods face significant difficulties. First, the problem is inherently NP-hard, and the solution space expands exponentially with the increasing number of users and satellites. Second, the high mobility of LEO satellites leads to frequent changes in network topology and system states, making the static or rule-based strategies ineffective for real-time decision-making. These limitations highlight the need for DRL approaches to optimize inter-satellite handover in ultra-dense, time-varying LEO satellite networks.

III. HRL-Based Inter-Satellite Handover Method

In this section, we first present the overall framework of the HRL-based inter-satellite handover approach. Then, we detail the Markov decision process (MDP) formulations and optimization methods for both temporal and spatial decision-making. Finally, we summarize the algorithmic procedure.

A. Overall Architecture

In dense LEO satellite constellations, a key challenge of user handover lies in jointly determining *when to initiate a handover* and *which satellite to handover to*. These two aspects, i.e., handover timing and handover target, are influenced by different factors across distinct dimensions. Therefore, we propose a HRL-based framework that explicitly decomposes the handover decision-making process into two hierarchical levels, as shown in Fig. 2.

- The temporal level operates at a coarser timescale, focusing on long-term planning. It uses a sequential policy that incorporates satellite mobility patterns, future load predictions, and inter-satellite interference modeling to determine when a handover should occur, thereby avoiding premature or unnecessary handovers.
- The spatial level operates at a finer timescale, focusing on short-term adaptation. Upon receiving the indication from the temporal level, it evaluates the visible satellite set by maximizing a utility function that jointly considers satellite load, signal quality, and remaining service time, thereby selecting the optimal handover target.

The HRL-based framework adopts a master–worker structure, where the temporal layer acts as a high-level controller responsible for deciding when to initiate a handover, while the spatial layer functions as an event-driven executor that determines the handover target only after a trigger is issued. By decoupling long-term timing decisions from short-term spatial selection, the two layers operate at different time scales with clearly defined decision scopes. Each decision-making layer can be independently designed, trained, and optimized. Meanwhile, coordination between the two layers is achieved through a hierarchical information exchange and feedback mechanism, in which the temporal layer constrains the execution scope of the spatial layer via handover triggering

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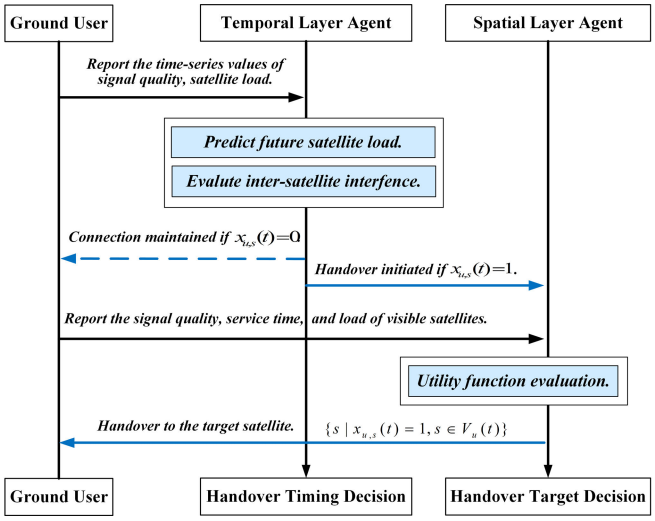


Fig. 2. The framework of HRL-based user handover process.

and handover candidates, while the execution outcomes of spatial decisions are implicitly fed back to the temporal layer through state transitions and reward signals [36].

B. The Handover Timing Decision-Making Process

As show in Fig. 3, the temporal module adopts a GRU-based actor-critic architecture and utilizes the PPO algorithm to optimize the handover timing. This module takes as input system state indicators with temporal correlations, including channel quality variations, historical load conditions, and service time trends, to decide whether a user should disconnect from the current satellite and initiate a handover. We model the temporal-level decision-making problem as a MDP, with the key elements defined as follows.

State: At each time slot t, the state is composed of both sequential and instantaneous features. The sequential features capture the historical dynamics of the system, while the instantaneous features reflect the current observable conditions. Accordingly, the overall state o_t^{temp} can be defined as

$$o_t^{temp} = \left[o_t^{seq}, o_t^{ins}\right]. \tag{16}$$

To effectively capture temporal dynamics, we incorporate the historical satellite load, signal quality, and remaining service time of the user u and its current serving satellite s into the sequential features representation, written by

$$o_t^{seq}(u,s) = \left[L_s, H_{u,s}, \Lambda_{u,s}\right](t-k\Delta t : t). \tag{17}$$

Here, $H_{u,s}$ denotes the SINR of the signal received by user u from satellite s, and $H_{u,s}$ represents the remaining service time that satellite s can provide to user u. Additionally, the instantaneous features, denoted as

$$o_t^{ins}(u,s) = \left[\theta_{u,s}(t), D_{u,s}(t), I_{u,s}(t)\right], \tag{18}$$

include information such as elevation angle, traffic demand, and interference. These sequential and instantaneous features are jointly fed as inputs into both the actor and critic networks.

Action: At each time slot t, the temporal decision is made independently for each user. Specifically, for user u, the temporal action is defined as a binary variable $a_u^{temp}(t) \in \{0,1\}$, where $a_u^{temp}(t)=1$ indicates that user u triggers a handover at time t, and $a_u^{temp}(t)=0$ means that the user maintains the current satellite connection. The actions of all users within the considered area are stacked into a vector form, represented as

$$a_t^{temp} = \left\{a_1^{temp}(t), a_2^{temp}(t), \ldots, a_U^{temp}(t)\right\}, \tag{19}$$

which determines whether each user in the target area should remain connected to its current serving satellite.

Reward: We first define a scoring function to quantify the QoS provided by the LEO satellite, based on the normalized satellite load $\bar{L}_s$, the normalized signal quality $\bar{H}_{u,s}$, and the normalized remaining service time $\bar{\Lambda}_{u,s}$. The QoS function is designed as

$$\Theta_t = \omega_1\left(1-\bar{L}_s(t)\right) + \omega_2\bar{H}_{u,s} + \omega_3\bar{\Lambda}_{u,s}, \tag{20}$$

where $\omega_1+\omega_2+\omega_3=1$ is satisfied.[2] Then, the reward for the temporal decision-making process is customized as

$$r_t^{temp} = \begin{cases} \textit{Case 1:} \quad \Delta\Theta, \ x_{u,s}(t)=0, \\ \textit{Case 2:} \quad \kappa\Delta\Theta, \ x_{u,s}(t)=1, \\ \textit{Case 3:} \quad \mu\Delta\Theta-\Omega, \ x_{u,s}(t)=1, \\ \textit{Case 4:} \quad -p, \ x_{u,s}(t)=1. \end{cases} \tag{21}$$

Specifically, the reward design is divided into the followings.

- *Case 1: The user does not perform a handover.* In this case, we simply evaluate the change in the QoS score.
- *Case 2: The user initiates a handover as the currently connected satellite is about to move out of the target coverage area, i.e., the elevation angle drops below a predefined threshold or the remaining service time falls within the safety margin.* The reward for this case is based on the change in the QoS score, with a scaling factor κ applied to encourage such behavior.
- *Case 3: The user performs a handover due to a significant performance degradation of the current satellite and the presence of a better alternative.* In this case, the reward is determined by both the QoS gain with a scaling factor μ and the associated handover cost, denoted by Ω.
- *Case 4: The user may perform handovers that are too early, too late, or occur too frequently.* Here, we introduce a penalty term $-p$ to discourage such behaviors.

In Eq. (21), the scaling factors κ and μ are introduced to distinguish necessary and unnecessary handovers, while Ω and p model the cost and penalty associated with handover operations. These parameters are designed to balance QoS improvement and handover stability, and their values are chosen to ensure that necessary handovers are encouraged, unnecessary handovers are penalized, and unjustified handovers are carefully controlled.

[2]To avoid subjective weight assignment, several representative weight configurations of the QoS function are evaluated in the simulation section. The results show that a relatively balanced configuration achieves better convergence than single-metric-dominated settings.

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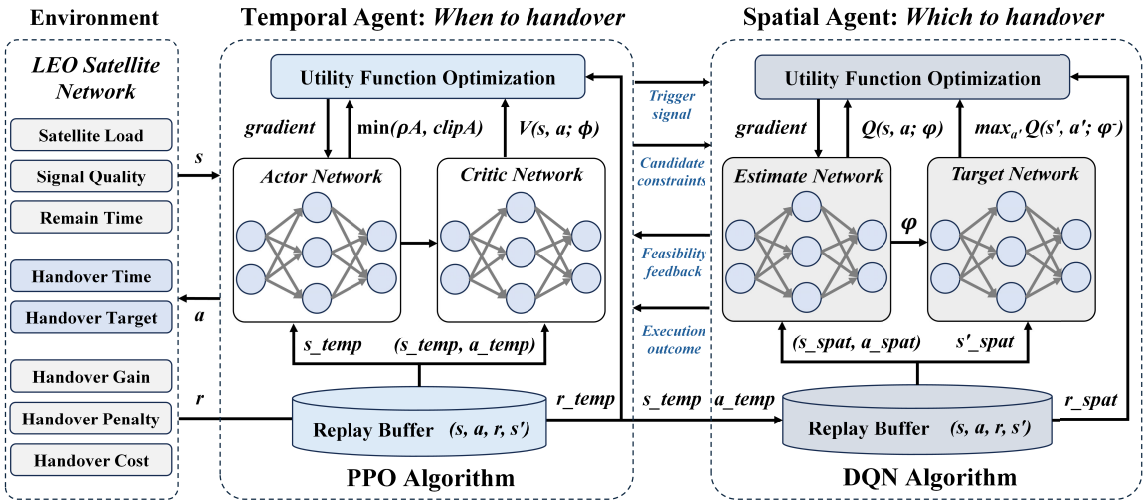


Fig. 3. The framework of HRL-based inter-satellite handover process.

To support temporal decision-making, we adopt an actor-critic framework that consists of a GRU-based actor network and a fully connected critic network.

Specifically, the historical sequential features $o_t^{seq}(u,s)$ are first fed into a GRU layer to extract the temporal dynamics. The GRU employs two gating mechanisms: the update gate $\mathbf{c_t}$ and the reset gate $\mathbf{e_t}$. The update process is defined as

$$\mathbf{c}_t = \sigma\left(\mathbf{W}_c\mathbf{o}_t + \mathbf{M}_c\mathbf{H}_{t-1}\right), \tag{22a}$$
$$\mathbf{e}_t = \sigma\left(\mathbf{W}_e\mathbf{o}_t + \mathbf{M}_e\mathbf{H}_{t-1}\right), \tag{22b}$$

where $\sigma(\cdot)$ represents the Sigmoid activation function. $\mathbf{o_t}$ is the input feature vector at time slot t, and $\mathbf{H_{t-1}}$ denotes the hidden state from the previous time slot. $\mathbf{W}$ and $\mathbf{M}$ are input and recurrent weight matrices, respectively. Next, the GRU layer computes the candidate hidden state and the updated hidden state, which are expressed as [37]

$$\tilde{\mathbf{H}}_t = \tanh\left(\mathbf{W}_h\mathbf{o}_t + \mathbf{M}_h\left(\mathbf{e}_t \odot \mathbf{H}_{t-1}\right)\right), \tag{23a}$$
$$\mathbf{H}_t = \left(1-\mathbf{c}_t\right)\odot\mathbf{H}_{t-1} + \mathbf{c}_t \odot \tilde{\mathbf{H}}_t. \tag{23b}$$

Here, $\tanh(\cdot)$ represents the Tanh activation function, and $\odot$ denotes the Hadamard product. $\tilde{\mathbf{H}}_\mathbf{t}$ is the candidate hidden state, and $\mathbf{H_t}$ is the final extracted temporal feature vector. It is worth noting that the GRU-based module does not assume perfectly accurate predictions of satellite load or interference. Instead, it serves as a temporal representation learner that extracts trend-level information from historical observations.

This hidden state is then concatenated with the instantaneous features, and the combined features are passed through fully connected layers to generate a probability distribution over possible actions. The output of the actor network is

$$\pi_\phi\left(a_t^{temp} \mid o_t^{temp}\right) = \mathrm{softmax}\left(\mathbf{W}_a\mathbf{o}_t + \mathbf{b}\right), \tag{24}$$

where $\mathbf{W}_a$ is the weight matrix, and $\mathbf{b}$ is the bias vector. The output gives the selection probability for each action. The critic network, implemented as a fully connected neural network, shares the same input as the actor network and outputs a scalar value $V_\phi(o_t^{temp})$. This value is used to compute the advantage function, represented by

$$A_t = G_{\pi_\phi}(o_t^{temp}, a_t^{temp}) - V_\phi(o_t^{temp}). \tag{25}$$

Here, G_{π_ϕ} is the expected cumulative return.

To ensure stable updates, the PPO algorithm adopts the importance sampling ratio [38], denoted by

$$\rho_t = \frac{\pi_\phi\left(a_t^{temp} \mid o_t^{temp}\right)}{\pi_{\phi_{\text{old}}}\left(a_t^{temp} \mid o_t^{temp}\right)}, \tag{26}$$

where π_ϕ is the current policy, and $\pi_{\phi\text{old}}$ is the old policy. The clipped loss function is defined as

$$\mathcal{J}_A = \mathbb{E}\left[\min\left(\rho_t A_t, \mathrm{clip}\left(\rho_t, 1-\epsilon, 1+\epsilon\right)A_t\right)\right], \tag{27}$$

where ϵ is a parameter for clipping. In addition to the policy loss, the PPO algorithm incorporates a value function loss and an entropy regularization term to improve prediction accuracy and encourage exploration, given by

$$\mathcal{J}_V = \mathbb{E}\left[\left(G_{\pi_\phi}(o_t^{temp}, a_t^{temp}) - V_\phi(o_t^{temp})\right)^2\right], \tag{28a}$$
$$\mathcal{J}_E = \mathbb{E}\left[\mathcal{H}\left(\pi_\phi\left(\cdot \mid o_t^{temp}\right)\right)\right]. \tag{28b}$$

$\mathcal{H}$ denotes the entropy of the policy distribution. The total loss combines all terms as

$$\mathcal{J}_\phi = -\mathcal{J}_A + c_1\cdot\mathcal{J}_V - c_2\cdot\mathcal{J}_E, \tag{29}$$

where c_1 and c_2 are the weighting coefficients for the value loss and entropy regularization, respectively.

C. The Handover Target Decision-Making Process

As show in Fig. 3, the spatial module is optimized based on the DQN algorithm to determine the optimal handover target. This module takes as input the information of all visible satellites except the currently connected one, including satellite load, signal quality and remaining service time. Moreover, the spatial module employs an experience replay and a target network to estimate the long-term service value associated with each visible satellite. Based on these value estimations,

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the module outputs a binary matrix that specifies whether each candidate satellite is selected for handover. The spatial-level decision-making problem is formulated as a MDP, with its key components defined as follows.

State: At each time slot t, the state of the spatial decision-making process is composed of two components: the information of the currently connected satellite and that of the candidate satellites, defined as

$$o_t^{spat} = \left[o_t^{conn}, o_t^{cand}\right]. \tag{30}$$

Specifically, the information available from the currently connected satellite s can be represented as

$$o_t^{conn}(u,s) = \left[\Upsilon_{u,s}(t), \theta_{u,s}(t), D_{u,s}(t), I_{u,s}(t)\right], \tag{31}$$

where $\Upsilon_{u,s}$ denotes the index of the satellite currently connected to the user u. In addition, the information associated with the candidate satellites can be expressed as

$$o_t^{cand}(u) = \left[L_{s'}(t), H_{u,s'}(t), \Lambda_{u,s'}(t)\right], s' \in \{\mathbf{V}_u/s\}, \tag{32}$$

which includes the load level, signal quality, and remaining service time of each candidate satellite s'.

Action: The action in the spatial decision-making process is defined as a binary vector matrix that represents the handover decisions between the users and the set of visible satellites, which is written as

$$a_t^{spat} = \begin{bmatrix} x_{1,1}(t) & x_{1,2}(t) & \cdots & x_{1,S}(t) \\ x_{2,1}(t) & x_{2,2}(t) & \cdots & x_{2,S}(t) \\ \vdots & \vdots & \ddots & \vdots \\ x_{U,1}(t) & x_{U,2}(t) & \cdots & x_{U,S}(t) \end{bmatrix}. \tag{33}$$

Here, $x_{u,s}(t) = 1$ indicates that user u handovers to satellite s, while $x_{u,s}(t) = 0$ means no handover to that satellite. To ensure the validity of handover decisions, a visibility-based masking mechanism is introduced, expressed by

$$\mathbf{m}_u(t) = \left[m_{u,1}(t), m_{u,2}(t), \ldots, m_{u,S}(t)\right], \tag{34}$$

where $m_{u,s}(t) = 1$ if satellite s is visible to user u at time slot t, and $m_{u,s}(t) = 0$ otherwise. During training, the mask is also applied to the action logits, represented by

$$\tilde{\zeta}_{u,s}(t) = \zeta_{u,s}(t) m_{u,s}(t) - \infty(1 - m_{u,s}(t)). \tag{35}$$

Here, $\zeta_{u,s}(t)$ represents the raw logit output, and $\tilde{\zeta}_{u,s}(t)$ is the resulting masked logit. Through a softmax function, the probability that user u selects satellite s' as the handover target is denoted as [39]

$$\chi_{u,s'}(t) = \frac{\exp\left(\tilde{\zeta}_{u,s'}(t)\right)}{\sum_{s=1}^{S} \exp\left(\tilde{\zeta}_{u,s}(t)\right)}. \tag{36}$$

Reward: To guide the selection of handover target, we construct a utility function that integrates multiple critical factors, as defined in Eq. (20), including signal quality, satellite load, and remaining service time. These factors reflect the trade-offs between short-term transmission performance and long-term service continuity. Accordingly, we design the reward as

$$r_t^{spat} = \Delta\Theta_t(u, s') - \Omega, \tag{37}$$

where Ω represents the handover cost.

To optimize the target selection strategy, we adopt the DQN algorithm as the spatial-level policy learning framework. The DQN algorithm enables the agent to estimate the state-action value function $Q(o_t^{spat}, a_t^{spat})$, and the Q-function is updated by approximating the Bellman equation [40], denoted by

$$Q(o_t, a_t) = \mathbb{E}\left[r_t + \gamma \max_{a_t'} Q\left(o_{t+1}, a_t'\right) \mid o_t, a_t\right]. \tag{38}$$

Here, γ is the discount factor. The DQN algorithm employs a neural network parameterized by φ to approximate the Q-function, and the network is trained by minimizing the mean squared error (MSE) between the predicted Q-value and the target Q-value. The loss function is expressed as

$$\mathcal{J}_\varphi = \frac{1}{K} \sum_{i=1}^{K} \left(Q_t(o_t^{spat}, a_t^{spat}; \varphi) - Q'_\varphi(i)\right)^2, \tag{39}$$

where K is the mini-batch size. This learning process enables the spatial agent to accurately evaluate the long-term utility of selecting each candidate satellite.

D. Algorithm Procedure

The overall procedure of the HRL-based inter-satellite handover method is summarized in Algorithm 1. It takes as input the user set $\mathbb{U}$, satellite set $\mathbb{S}$, visibility matrix, policy networks, and hyperparameters, and initializes the temporal and spatial policy parameters, the replay buffer, and the environment state. At the beginning of each training episode, the environment is reset. For each time step, the temporal agent observes sequential features o_t^{seq} and instantaneous features o_t^{ins}, which are then combined to form the temporal observation o_t^{temp}. Next, the GRU-based temporal policy network π_ϕ generates an action a_t^{temp} that determines whether a handover should be initiated. If a handover is triggered, the algorithm proceeds to the spatial decision phase. The spatial decision module collects features of connected and candidate satellites to form the spatial state o_t^{spat}. Then, the spatial agent filters out invisible satellites using a visibility mask, and selects the target satellite a_t^{spat} based on computed selection probabilities. The handover is executed, and the corresponding spatial reward r_t^{spat} is obtained. If no handover is initiated, the temporal agent evaluates the current QoS score Θ_t and receives a temporal reward r_t^{temp}. It should be noted that after the spatial layer selects and executes a handover target, the resulting QoS variation, handover cost, and service stability are reflected in the next system state and reward. These updated observations are then used by the temporal layer in subsequent decision-making and policy updates. However, in scenarios where the spatial layer fails to identify a target satellite with sufficient QoS gain or acceptable handover cost, the corresponding outcome leads to a low or negative reward. This penalty is propagated to the temporal layer through the reward function, discouraging premature or infeasible handover triggering under similar conditions. Therefore, the temporal layer can adaptively learn to postpone or suppress handovers without introducing explicit rule-based rollback mechanisms. The effective step-size ratio between the temporal and spatial layers satisfies $\Delta T_{\text{temp}} \gg \Delta T_{\text{spat}}$.

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Algorithm 1 HRL-Based Inter-Satellite Handover Algorithm

Input: User set $\mathbb{U}$, LEO satellite set $\mathbb{S}$, visibility matrix, policy networks and hyperparameters.

Output: Temporal decision a^{temp}, spatial decision a^{spat}, and the trained policies of inter-satellite handover.

Initialize:
Temporal policy parameters ϕ, spatial policy parameters φ, replay buffer and environment state.
for episode = 1 to N_{ep} **do**
  Reset environment of the LEO satellite network;
  for $t = 1$ to T **do**
    Observe sequential features o_t^{seq} in Eq. (17);
    Observe instantaneous features o_t^{ins} in Eq. (18);
    Combine into state o_t^{temp} in Eq. (16);
    Use GRU-based actor π_ϕ to compute action based on Eq. (24) and execute action a_t^{temp} in Eq. (19);
    Compute QoS score Θ_t in Eq. (20) and determine reward r_t^{temp} based on Eq. (21);
    if $a_t^{temp} == 1$ **then**
      Observe connected satellites o_t^{conn} in Eq. (31);
      Observe candidate satellites o_t^{cand} in Eq. (32);
      Form spatial state o_t^{spat} in Eq. (30);
      Mask invisible satellites with $\mathbf{m}_u(t)$ and compute selection probabilities based on Eq. (34)–(36);
      Choose handover target a_t^{spat} based on Eq. (33);
      Execute handover behavior and compute reward r_t^{spat} based on Eq. (37);
    end if
  end for
  Store transition (o_t, a_t, r_t, o_{t+1}) into buffer;
  if PPO Update **then**
    Compute advantage A_t in Eq. (25);
    Compute policy loss $\mathcal{J}_A$ in Eq. (27), and compute value loss $\mathcal{J}_V$ and entropy loss $\mathcal{J}_E$ in Eq. (28);
    Update temporal policy π_ϕ based on Eq. (29);
  end if
  if DQN Update **then**
    Sample mini-batch of transitions from buffer;
    Compute target Q value and update Q-network by minimizing MSE loss based on Eq. (39);
    Periodically update $Q'_\varphi \leftarrow Q_\varphi$;
  end if
end for

Then, each transition tuple (o_t, a_t, r_t, o_{t+1}) is stored in the replay buffer. For temporal policy updates, the PPO algorithm is employed to update the temporal policy parameters ϕ. For the spatial decision-making layer, the DQN algorithm is adopted to update the Q-network parameters φ. During training, the PPO-based temporal policy is updated periodically using accumulated transitions, while the DQN-based spatial policy is updated more frequently in an event-driven manner. It is worth noting that the temporal and spatial policies are trained asynchronously and do not share parameters directly. Their coordination is achieved through environment-mediated feedback, where the spatial execution outcome influences the temporal policy only via reward shaping and state evolution, ensuring stable hierarchical learning. As a result, the proposed HRL-based framework achieves coordinated optimization of handover triggering and target selection, leading to more robust handover decisions and improved service quality under dynamic topology and interference conditions.

The complexity analysis of the proposed HRL method is given as follows. During training execution, the temporal agent employs a PPO algorithm with a GRU-based actor–critic architecture. Its training complexity is dominated by the forward and backward propagation of the GRU and fully connected layers, which scales as $\mathcal{O}(N_{\text{temp}})$, where N_{temp} denotes the number of temporal network parameters and is independent of the number of visible satellites. The spatial agent adopts a DQN framework. Its training complexity scales linearly with the number of candidate satellites $V_u(t)$, i.e., $\mathcal{O}(V_u(t) \cdot N_{\text{spat}})$, where N_{spat} is the number of parameters in the Q-network. During online execution, the temporal agent performs a lightweight binary decision at each time step, resulting in a constant inference cost $\mathcal{O}(1)$. The spatial agent is activated only when a handover is triggered. Therefore, the expected inference complexity per time step can be expressed as $\mathcal{O}(1) + \mathbb{P}_{\text{HO}} \cdot \mathcal{O}(|V_u(t)|)$, where $\mathbb{P}_{\text{HO}}$ denotes the handover triggering probability. Since unnecessary handovers are explicitly suppressed by the temporal policy, $\mathbb{P}_{\text{HO}}$ remains low in practice, leading to substantial computational savings in dense LEO constellation scenarios.

IV. Simulation Results

A. Simulation Settings

We first built a simulation environment using the system tool kit (STK), incorporating the parameters of the Starlink constellation. The algorithm was implemented in Python and integrated into the simulation framework. The LEO constellation consists of 1584 satellites distributed over 72 orbital planes, with 22 satellites per plane. The satellites operate at an orbital altitude of 550 km with a velocity of 7.5 km/s and an inclination angle of 53°. A minimum elevation angle of 25° was configured to determine satellite visibility from the ground user. The simulation was conducted over a geographic region defined by the latitude range of $[30°\text{N}, 32°\text{N}]$ and longitude range of $[116°\text{E}, 118°\text{E}]$, which is used to emulate a regional service hotspot under dense satellite coverage. Within this region, we deployed 100 users, and user traffic arrival follows a Poisson process with an average rate of $\lambda = 0.1\ \text{s}^{-1}$. Moreover, each user can dynamically observe 3–6 satellites simultaneously at any given moment, and this number further increases in highly overlapped coverage regions. It should be emphasized that the simulation duration is set to 10 s, which is sufficient to cover multiple satellite visibility windows and capture the short-term orbital dynamics that dominate inter-satellite handover behavior in LEO satellite systems. The key simulation parameters is summarized in Table I.

The HRL-based handover algorithm is designed with two levels. At the temporal level, a PPO algorithm combined with a GRU-based actor-critic network is employed. The GRU-based actor network consists of a three-layer stacked GRU structure, with the first two layers having 256 hidden units each, and the

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TABLE I
SIMULATION PARAMETERS

Simulation Parameter	Value
Number of LEO Satellite	1584
Number of LEO Orbits	72
Number of Satellites per Orbits	22
Orbit Altitude (km)	550
Orbit Inclination (°)	53
Satellite Speed (km/s)	7.5
Minimum Visible Elevation Angle (°)	25
Satellite Capacity (Gbps)	15
Number of Users	100
Traffic Arrival Rate (s^{-1})	0.1
Carrier Frequency (GHz)	12
System Bandwidth (MHz)	500
Transmit Power (dBW)	20
Rician factor (dB)	10
Shadowing variance (dB)	6
Satellite Antenna Gain (dBi)	32
User Antenna Gain (dBi)	35
Noise Power Density (dBm/Hz)	-174
Simulation Time (min)	10

third layer having 128 hidden units. It processes a historical window of 10 time slots covering satellite load, signal quality, and service time. The features processed by the GRU layers are then fed into a two-layer fully connected hidden layers with 128 and 64 neurons respectively, using ReLU activation. The critic network consists of three fully connected hidden layers with 256, 128, and 64 neurons, respectively. The PPO algorithm is configured with a learning rate $\xi_{temp} = 3 \times 10^{-4}$, a discount factor $\gamma = 0.99$, a clipping parameter $\epsilon_{temp} = 0.2$, and Adam optimizer. At the spatial level, a DQN algorithm is adopted. The DQN network consists of four fully connected hidden layers with 256, 128, 128, and 64 neurons, respectively. The learning rate $\alpha_{spat} = 1 \times 10^{-3}$ is used along with an ϵ-greedy exploration strategy where the initial exploration probability is set to 1.0 and decays linearly at a rate of 0.995 until a minimum of 0.01 is reached during training.

In this paper, we compare our proposed method (MO-HRL) with four baseline schemes:

- Multi-objective RL-based Handover (MO-RL): This approach jointly optimizes multiple objectives, including satellite load, signal quality, and remaining service time, using a non-hierarchical RL framework to optimize.
- Satellite Load-driven RL Handover (SL-RL): This baseline considers only the satellite load as the criterion, and applies RL to determine the handover strategy.
- Signal Quality-driven RL Handover (SQ-RL): This approach learns the handover strategy using RL based on the received signal quality of users.
- Service Time-driven RL Handover (ST-RL): This method takes the remaining service time as the guiding metric and learns handover strategies through RL.

To ensure a fair comparison, the implementation details of all baseline algorithms are clarified as follows. The MO-RL scheme adopts a non-hierarchical, single-agent reinforcement learning framework, in which the state representation, action space and reward function are exactly the same as those of the HRL approach. For the single-factor baselines, namely SL-RL, SQ-RL, and ST-RL, the same state representation and action space are retained to ensure consistency. However, their reward functions are simplified to focus on a single dominant metric. The comparison results between our proposed algorithm and the baseline methods will be analyzed in the following.

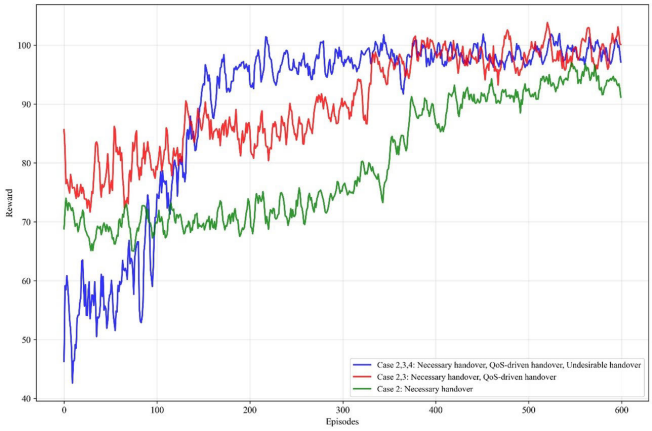


Fig. 4. The convergence performance of the temporal decision-making process for handover timing optimization under different reward settings.

B. Convergence Performance Analysis

To evaluate the convergence performance of the proposed HRL-based inter-satellite handover algorithm, we conduct experiments under various reward settings for both the temporal and spatial decision-making processes.

As illustrated in Fig. 4, we evaluate the convergence performance of the temporal-level decision-making process under three reward configurations: I) *Case 2*, which considers only necessary handovers; II) *Case 2,3*, which additionally rewards QoS-driven handovers; and III) *Case 2,3,4*, which further penalizes undesirable handovers. It is worth noting that the values of parameter κ, μ amd Ω, remain unchanged across different reward configurations, and the observed performance gains mainly stem from the incorporation of QoS incentives and handover penalties. The results show that refining the reward structure leads to significant improvements in both convergence speed and final performance. The blue curve quickly improves after approximately 150 episodes and stabilizes after 300 episodes, eventually achieving a reward of over 102. In contrast, the red curve converges to around 99, while the green curve reaches only about 92 and converges more slowly. These results highlight that incorporating QoS incentives and penalizing undesirable handovers enables the agent to learn more effective and stable policies, with a reward improvement of approximately 10% over the other cases.

Fig. 5 presents the convergence results of the spatial-level decision-making process under different reward weight configurations for satellite load, signal quality, and remaining service time. Among the four configurations tested, the reward setting with weights of (0.4, 0.3, 0.3), which adopts a more

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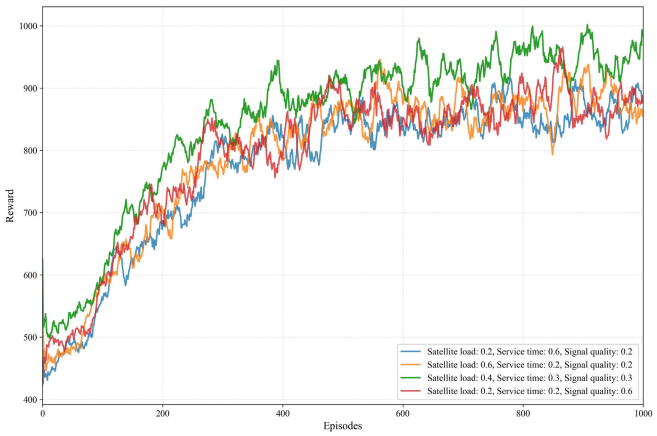


Fig. 5. The convergence performance of the spatial decision-making process for handover target selection under different reward settings.

balanced distribution across the three objectives, exhibits the best convergence performance of a final reward of approximately 990. In contrast, the other three configurations, which emphasize a single factor, such as satellite load (0.6, 0.2, 0.2), service time (0.2, 0.6, 0.2), or signal quality (0.2, 0.2, 0.6), converge more slowly and reach lower final rewards of around 860, 890, and 880, respectively. The configurations dominated by a single metric converge more slowly and reach lower final rewards. These results indicate that overly emphasizing a single objective may limit decision quality, whereas a balanced reward better captures the trade-offs among signal quality, load balancing, and service continuity. Although a balanced configuration is adopted in this paper, the proposed framework is not restricted to a fixed weight setting and can accommodate different QoS preferences.

C. Comparison of Handover Performance

In this section, we compare the handover success rate and handover frequency under varying conditions of signal quality, satellite load, and remaining service time.

As shown in Fig. 6a–Fig. 6c, the MO-HRL algorithm consistently outperforms all baseline methods in terms of handover success rate. In Fig. 6a, as the normalized signal quality increases from 0.3 to 1.0, all methods show an upward trend. The MO-HRL approach achieves the highest success rate, increasing from around 0.81 to nearly 0.99. The MO-RL algorithm follows with a success rate ranging from 0.77 to 0.95. The three single-factor RL methods (SQ-RL, SL-RL, ST-RL) exhibit relatively lower performance and converge near 0.86 at best. This demonstrates that incorporating multiple decision factors and using a hierarchical structure significantly enhances the ability to identify valid handover opportunities, particularly showing significant advantages under good channel conditions. In Fig. 6b, with increasing satellite load, the handover success rate of all methods declines. However, the MO-HRL method still maintains a relatively high performance, decreasing from approximately 0.95 to 0.80, while the MO-RL scheme drops from 0.92 to 0.75. The other methods fall below 0.70 when the load reaches 0.8. This shows that our proposed method is more robust under heavy-load conditions, as it can better balance traffic distribution during handover. In Fig. 6c, when normalized remaining service time increases, the success rate of all schemes improves. The MO-HRL algorithm maintains a significant advantage, achieving nearly 0.99 when the remaining service time approaches 1.0. This indicates its superior capability in learning from temporal usage patterns, enabling more stable and persistent connection strategies.

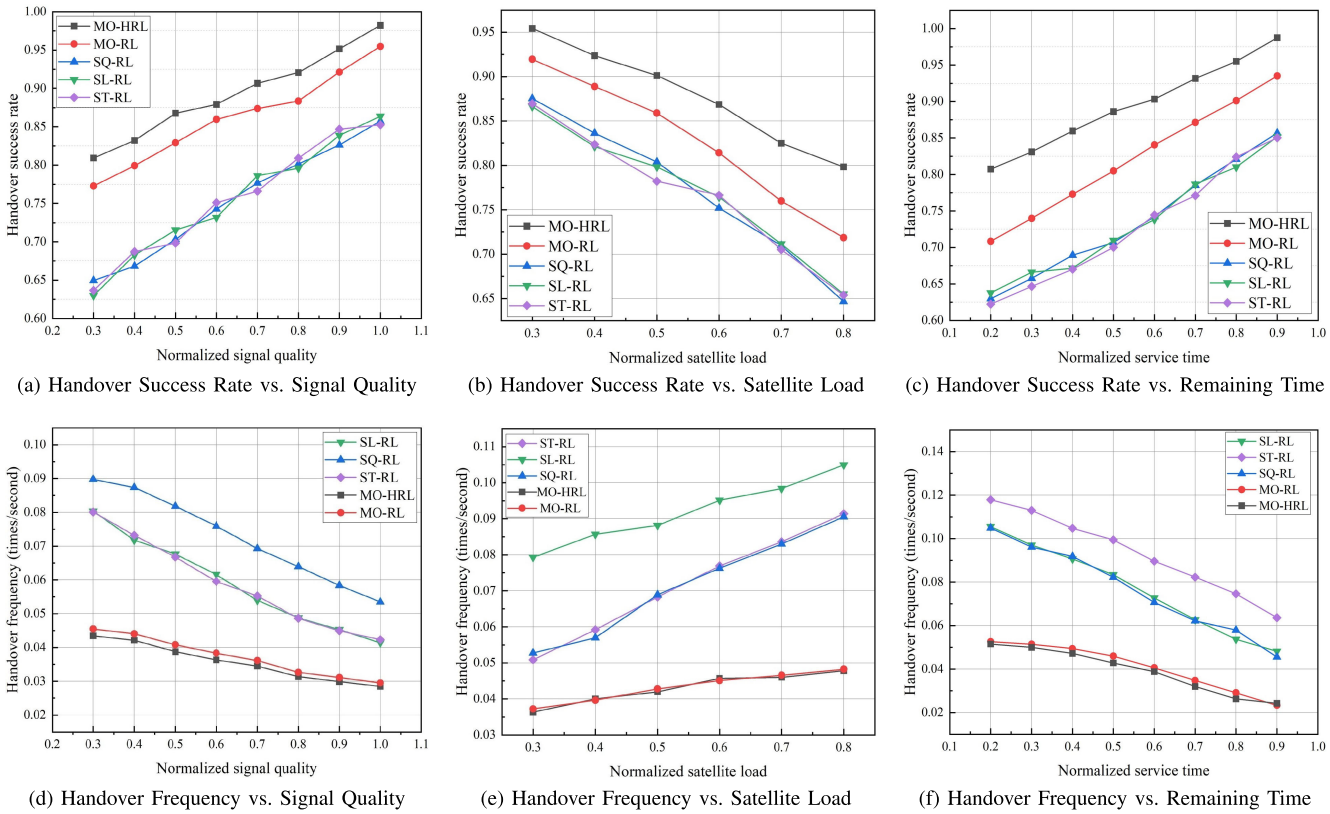
Fig. 6d–Fig. 6f evaluate the handover frequency of each method under the same varying conditions, showing the efficiency and stability of handover decisions. In Fig. 6d, as the signal quality improves, handover frequency for all methods decreases. The MO-HRL and MO-RL algorithms maintain the lowest frequency throughout, dropping from approximately 0.045 to 0.025 times per second. In contrast, the SQ-RL algorithm shows the highest frequency, above 0.09 at low signal quality and decreasing slowly. This indicates that the single-factor strategy focused on signal quality is prone to unnecessary frequent handovers, especially under fluctuating signal conditions. In Fig. 6e, with increasing satellite load, the handover frequency of SL-RL rises steeply from 0.08 to above 0.105, indicating high sensitivity and instability under satellite load pressure. Meanwhile, the MO-HRL approach maintains a relatively stable and low frequency, increasing only slightly from 0.037 to 0.045, showing its advantage in managing traffic fluctuations efficiently. In Fig. 6f, the longer remaining service times lead to fewer handovers across all methods. The MO-HRL scheme achieves the most stable performance, with frequency decreasing to below 0.025. However, the ST-RL approach remains the most sensitive to service time, with frequency dropping from 0.12 to 0.06, suggesting that relying on service duration without considering load or QoS may lead to excessive switching when user sessions are short.

D. Comparison of System Performance

To verify the effectiveness and robustness of the proposed approach under different geographic latitude conditions, we consider three representative regions, including low-latitude, mid-latitude, and mid-to-high-latitude areas. The performance of different handover algorithms is evaluated and compared in terms of handover success rate and handover frequency.

Fig. 7 illustrates the handover success rate of different algorithms across low-latitude $[0°\mathrm{N}, 2°\mathrm{N}]$, mid-latitude $[30°\mathrm{N}, 32°\mathrm{N}]$, and mid-to-high-latitude $[45°\mathrm{N}, 47°\mathrm{N}]$ regions. An overall increasing trend with latitude can be observed for all schemes, which is mainly attributed to enhanced multi-orbit coverage and a larger candidate satellite set at higher latitudes. MO-HRL consistently achieves the highest success rate, increasing from approximately 0.95 at low latitudes to about 0.98 at mid latitudes, and further approaching 0.99 at mid-to-high latitudes. The MO-RL baseline follows a similar trend but remains lower, with values around 0.91, 0.95, and 0.97, respectively. In contrast, the single-objective baselines achieve only about 0.81–0.82 at low latitudes and improve to around 0.88–0.89 at higher latitudes. These results demonstrate that MO-HRL maintains a consistently high and stable handover success rate across different geographic regions, highlighting

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(a) Handover Success Rate vs. Signal Quality (b) Handover Success Rate vs. Satellite Load (c) Handover Success Rate vs. Remaining Time

(d) Handover Frequency vs. Signal Quality (e) Handover Frequency vs. Satellite Load (f) Handover Frequency vs. Remaining Time

Fig. 6. The handover performance of different algorithms under different influencing factors.

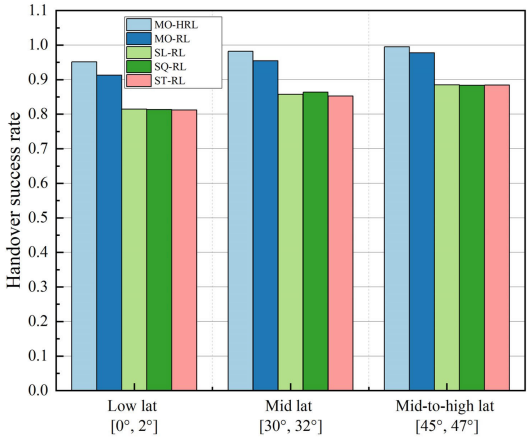


Fig. 7. The comparison of handover success rate of different algorithms in low, mid, mid-to-high latitude regions.

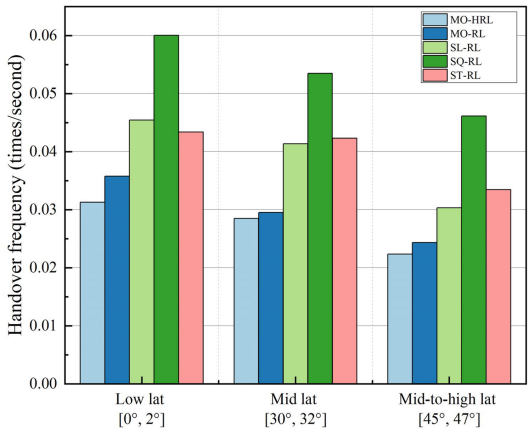


Fig. 8. The comparison of handover frequency of different algorithms in low, mid, mid-to-high latitude regions.

its superior robustness and adaptability under complex multi-satellite visibility and interference conditions.

Fig. 8 compares the handover frequency of different algorithms across latitude regions. It can be clearly observed that handover frequency decreases with increasing latitude for all schemes, indicating that improved coverage continuity and better satellite visibility at higher latitudes reduce the need for frequent handovers. MO-HRL consistently exhibits the lowest handover frequency, decreasing from approximately 0.031 times/s at low latitudes to about 0.029 times/s at mid latitudes, and further to around 0.022 times/s at mid-to-high latitudes. MO-RL shows slightly higher values, at roughly 0.036, 0.030, and 0.024 times/s, respectively. In contrast, SQ-RL results in the most frequent handovers, reaching nearly 0.060 times/s at low latitudes and remaining as high as 0.046 times/s even at higher latitudes. These results indicate that MO-HRL effectively suppresses unnecessary and excessive handovers while maintaining high success rates, achieving a balance between handover stability and service continuity.

To further evaluate the effectiveness of different handover strategies under varying satellite visibility conditions, we analyze two key performance metrics: transmission rate of users and satellite load balancing degree of system.

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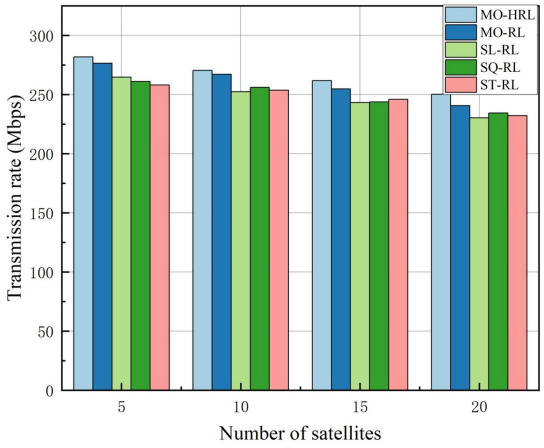


Fig. 9. The comparison of transmission rates under different numbers of visible satellites.

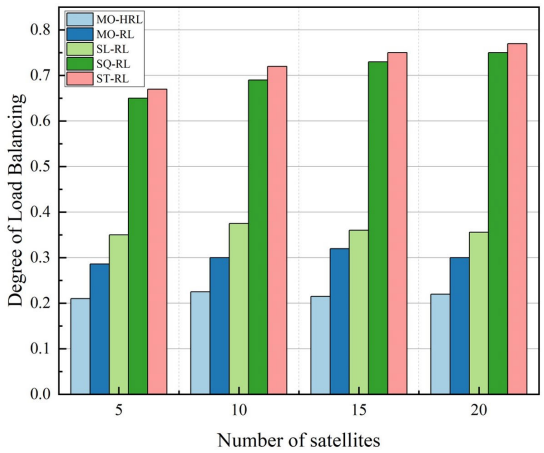


Fig. 10. The comparison of satellite load balancing under different numbers of visible satellites.

As presented in Fig. 9, the transmission rate of all methods declines as the number of visible satellites increases. This overall downward trend is caused by the increased interference and competition for limited spectrum and beam resources. The proposed MO-HRL algorithm consistently achieves the highest transmission rate across all visibility levels. As the number of satellites increases to 20, the rates for all methods drop, with the MO-HRL algorithm still maintaining around 250 Mbps, while the worst-performing method (ST-RL) falls to approximately 235 Mbps. These results demonstrate that the design of MO-HRL enables more effective interference avoidance and resource utilization, thereby maintaining better throughput under dense satellite conditions.

Fig. 10 provides the degree of load balancing under different numbers of visible satellites. It is important to note that a lower value of this metric indicates better load balancing, i.e., a more even distribution of users across LEO satellites and more efficient utilization of system resources. As the number of visible satellites increases, the load balancing degree of the single-factor RL approach tends to grow, reflecting their limited ability to coordinate user distribution and avoid satellite congestion. In contrast, MO-HRL leverages hierarchical decision-making and multi-objective optimization to achieve a more reasonable trade-off between service quality and resource utilization. Even as the number of visible satellites increases and the scheduling complexity intensifies, its advantage in load distribution remains consistently stable.

V. Conclusion

In this paper, we investigated the inter-satellite handover problem in dense LEO satellite constellations, with the goal of maximizing system throughput, minimizing unnecessary handovers, and achieving load balancing. A key challenge lies in the tight coupling between handover timing and handover target. An inappropriate handover time may limit the availability of optimal target satellites, while the quality of the target selection relies on the timing of the handover. To address the spatio-temporal coupling, we proposed a HRL-based framework that decouples the decision-making into two sub-tasks. Specifically, a temporal agent was designed to determine when to trigger a handover by forecasting future satellite load and interference trends based on the PPO algorithm. Meanwhile, a spatial agent utilizing the DQN algorithm was responsible for selecting the most appropriate target satellite, guided by a utility function that incorporates signal quality, service continuity, and load conditions. Extensive experiments were conducted and showed that our proposed handover strategy significantly outperforms traditional and non-hierarchical DRL approaches. In future work, we plan to extend our framework by considering satellite beamforming dynamics, aiming to develop more adaptive handover strategies in response to highly dynamic LEO satellite networks.

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